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TWO OBSERVABLE A AND B ARE SAID TO BE **COMPLEMENTARY** IF THERE CAN BE NO SINGLE BASIC MEASUREMENT THAT PROVIDES US WITH FULL INFORMATION ABOUT BOTH A AND B

TWO OBSERVABLE A AND B ARE SAID TO BE **COMPATIBLE** IF THERE IS A BASIS MEASUREMENT THAT CAN DETERMINE THE VALUES OF BOTH A AND B TOGETHER

A AND B COMMUTE (AND $[A, B] = 0$) $\Leftrightarrow$ A AND B ARE **COMPATIBLE**

PROVE THIS $\leftarrow$

$[A, B] = 0 \Leftrightarrow A$ AND B ARE **COMPATIBLE**



1° " $\Leftarrow$ "

I CAN HAVE A SET OF EIGENSTATES OF A

$$\left. \begin{aligned} \{ |h\rangle \} \quad A &= \sum_n a_n |h\rangle \langle h| \\ B &= \sum_n b_n |h\rangle \langle h| \end{aligned} \right\} \text{SINCE ARE COMPATIBLE}$$

CALCULATE $[A, B]$

$$[A, B] = AB - BA = \left(\sum_n a_n |h\rangle \langle h| \right) \left(\sum_m b_m |m\rangle \langle m| \right) - \left(\sum_m b_m |m\rangle \langle m| \right) \left(\sum_n a_n |h\rangle \langle h| \right) =$$

$$= \sum_{n,m} a_n b_m |h\rangle \langle h| |m\rangle \langle m| - \sum_{n,m} b_m a_n |m\rangle \langle m| |h\rangle \langle h|$$

$\langle n | m \rangle = \delta_{nm}$

$$= \sum_{n,m} a_n b_m |h\rangle \langle h| \delta_{nm} - \sum_{n,m} b_m a_n |m\rangle \langle m| \delta_{mn} =$$

$$= \sum_n a_n b_n |h\rangle \langle h| - \sum_n b_n a_n |h\rangle \langle h| = \boxed{0}$$



2° " $\Rightarrow$ "

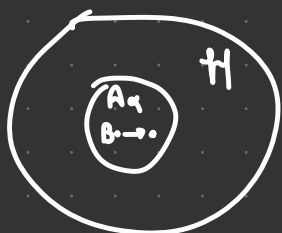
$$[A, B] = 0$$

$$A|\psi\rangle = A_d|\psi\rangle$$

$$B|\psi\rangle = ?$$

$$A(B|\psi\rangle) = AB|\psi\rangle = BA|\psi\rangle = BA_d|\psi\rangle = A_d(B|\psi\rangle)$$

So $B|\psi\rangle$ IS ALSO AN EIGENVECTOR OF A



$$B|b_a\rangle = B_d|b_d\rangle$$

$\rightarrow$ same

S_x, S_y AND S_z

PROVE THAT $[S_x, S_y] = i\hbar S_z$

$\hookrightarrow$ AND S_x AND S_y DO NOT COMMUTE SINCE $i\hbar S_z \neq 0$

$\{ |z+\rangle, |z-\rangle \}$

$$S_x = \frac{\hbar}{2} X \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$S_y = \frac{\hbar}{2} Y \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$S_z = \frac{\hbar}{2} Z \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\left[\frac{\hbar}{2} X, \frac{\hbar}{2} Y \right] = i\hbar \frac{\hbar}{2} Z$$

$$\frac{\hbar^2}{4} [X, Y] = i\hbar \frac{\hbar}{2} Z \rightarrow [X, Y] = 2iZ \rightarrow$$

$$\rightarrow XY - YX = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} - \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$XY - YX = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} - \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} 2i & 0 \\ 0 & -2i \end{bmatrix} = 2i \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 2iZ$$

SUMMARIZING

QUANTUM STATE

QUANTUM STATES

QUANTUM MEASUREMENT

INFORMATION AND QUANTUM SYSTEM



SIGNAL	
$ z+\rangle$	BAITHS BY <u>CAND</u>
$ z-\rangle$	BAITHS BY <u>SEA</u>
$ x+\rangle$	BAITHS <u>NOT COMING</u>

WE CONSIDER 3 DIFFERENT QUANTUM STATES

INFORMATION AND QUANTUM SYSTEM

SIGNAL	
$ z+\rangle$	D_1
$ z-\rangle$	D_2
$ x+\rangle$	

D_1 CLICKS $\rightarrow$ EITHER CAND OR NOT AT ALL (DEFINITELY NOT SEA)

D_2 CLICKS $\rightarrow$ EITHER BAITHS SEA OR NOT AT ALL (DEFINITELY NOT CAND)

NOT HELPFUL !!

CHANGE

SIGNAL	
$ z+\rangle$	D_1
$ z-\rangle$	D_2
$ x+\rangle$	

D_1 CLICKS $\rightarrow$ CAN NOT CONCLUDE ANYTHING

D_2 CLICKS $\rightarrow$ EITHER SEA OR CAND

NOT HELPFUL !!

INFORMATION: ABILITY TO DISTINGUISH RELIABLY BETWEEN POSSIBLE ALTERNATIVES

$\Rightarrow$ 1 BIT

IF NON-ORTHOGONAL QUANTUM SIGNAL (STATES) ARE USED, THEN INFORMATION CANNOT BE PERFECTLY RETRIEVED

WE DEFINE THE INFORMATION CAPACITY OF A QUANTUM SYSTEM AS THE MAXIMUM NUMBER OF BITS THAT CAN BE STORED AND RELIABLY READ BY A MEASUREMENT

THE INFORMATION CAPACITY OF A QUANTUM SYSTEM DESCRIBED BY A HILBERT SPACE OF DIMENSION d IS $\log_2(d)$

$\rightarrow$ WHAT HAPPENS IF WE TRY TO ENCODE A QUANTITY OF INFORMATION THAT EXCEEDS THE INFORMATION CAPACITY?

Basic decoding theorem

If Alice encodes N equally likely message values as states in a quantum system with $\dim(\mathcal{H}) = d$, and if Bob decodes this by performing a basic measurement (perhaps on a larger Hilbert space) and inferring the message from the result, Bob's probability of error is bounded by

$$P_E \geq 1 - \frac{d}{N}$$

or, equivalently, the success probability is $P_S \leq \min \left[\frac{d}{N}, 1 \right]$

SIGNAL MESSAGE VALUE

$|z_+\rangle$

LAND

$d=2 \quad N=3$

$|z_-\rangle$

SEA

$P_E \geq 1 - \frac{2}{3} = 0.334$

$|x_+\rangle$

NOT COMING

↳ THE BEST BOB CAN DO IS TO GUESS THE CORRECT MESSAGE VALUE 2/3 OF THE TIMES

BOB NEED TO RELIABLY DISTINGUISH BETWEEN THE POSSIBLE SIGNALS (STATES)

ONE STATE HILBERT SPACE $\mathcal{H}$ OF DIMENSION 2

ASSUME THE STATE $|\alpha_0\rangle$ AND $|\alpha_1\rangle$



← PICTORIAL REPRESENTATION

① ORTHOGONAL $\Rightarrow \langle \alpha_0 | \alpha_1 \rangle = 0$

↳ BEST CASE FOR BOB $|\alpha_0\rangle = |\alpha_0\rangle$

$|\alpha_1\rangle = |\alpha_1\rangle$. SINCE THE STATES ARE ORTHOGONAL $P_S = 1$

② OPTION 2 $|\alpha_0\rangle$ AND $|\alpha_1\rangle$ NOT ORTHOGONAL

Basic distinguishability theorem

For two equally likely states $|\alpha_0\rangle$ and $|\alpha_1\rangle$ such that $|\langle \alpha_0 | \alpha_1 \rangle| = \cos \theta$, the probability P_S of correctly identifying the state by a basic measurement is bounded by

$$P_S \leq \frac{1}{2}(1 + \sin \theta)$$

Equality is achievable by a particular choice of observable.

IN GENERAL ONLY ORTHOGONAL QUANTUM STATES OF A SYSTEM ARE COMPLETELY DISTINGUISHABLE BY MEASUREMENT

MEASUREMENT

$|\psi\rangle$ STATE

A OBSERVABLE

$\langle A \rangle = \langle \psi | A | \psi \rangle$ EXPECTATION VALUE

$\langle \Delta A^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2$ VARIANCE

$$\Delta A = \sqrt{\langle \Delta A^2 \rangle}$$

TWO OBSERVABLES A AND B ARE SAID TO BE COMPONENTARY //

SUPPOSE WE HAVE TWO OBSERVABLES A AND B FOR A QUANTUM SYSTEM
CAN WE SAY ANYTHING ABOUT THE UNCERTAINTY OF THE RESULT WHEN WE PERFORM A MEASUREMENT //

WE DEFINE THE OPERATOR
 $\Delta A = A - \langle A \rangle$
 $\Delta B = B - \langle B \rangle$
PROPERTY $\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$
 $\Delta A = \langle \psi | \Delta A | \psi \rangle$
 $\Delta B = \langle \psi | \Delta B | \psi \rangle$

Q20.

$$\langle \Delta_A A \rangle = \langle \psi | (A - \langle A \rangle) | \psi \rangle = \langle \psi | A | \psi \rangle - \langle A \rangle \langle \psi | \psi \rangle$$

$\Rightarrow$

$$\text{one can verify that } \langle \psi | (\Delta_A A)^2 | \psi \rangle = \langle \Delta A^2 \rangle \quad \langle \psi | (\Delta_B B | \psi \rangle = \langle \Delta B^2 \rangle$$

we introduce the two vectors

$$|a\rangle = \Delta_A A | \psi \rangle \quad |b\rangle = \Delta_B B | \psi \rangle$$

note that $|a\rangle$ 

THE UNCERTAINTY RELATION

SCHWARTZ INEQUALITY: if $|a\rangle$ and $|b\rangle$ are two vectors of a Hilbert space $\langle a | a \rangle \langle b | b \rangle \geq |\langle a | b \rangle|^2$

$$|a\rangle = \Delta_A A | \psi \rangle \quad |b\rangle = \Delta_B B | \psi \rangle$$

$\Downarrow$

$$\langle \psi | (\Delta_A A)^2 | \psi \rangle \langle \psi | (\Delta_B B)^2 | \psi \rangle \geq \langle \psi | \dots \rangle$$

$$F_G = \frac{1}{2} [F, G] + \frac{1}{2} \{F, G\}$$

ANTICOMMUTATOR

$$\{F, G\} = FG + GF$$

$$\textcircled{F_G} = \frac{1}{2} [F, G] + \frac{1}{2} \{F, G\}$$

$$\{F, G\} = FG + GF$$

$$= \frac{1}{2} (FG - GF) + \frac{1}{2} (FG + GF) = \textcircled{FG}$$

$$[F, G]^{\dagger} = (FG - GF)^{\dagger} = G^{\dagger} F^{\dagger} - F^{\dagger} G^{\dagger} \quad (\text{REMEMBER: } (AB)^{\dagger} = B^{\dagger} A^{\dagger})$$

LET ASSUME $F = F^{\dagger}$ AND $G = G^{\dagger}$ (BOTH HERMITIAN)

$$\Rightarrow [F, G]^{\dagger} = GF - FG = [G, F] = -[F, G] \quad \text{ANTIHHERMITIAN}$$

$$\{F, G\}^{\dagger} = GF + FG = \{F, G\} \quad \text{HERMITIAN}$$

PART 2

$$\left[\begin{array}{l} [F, G] \text{ is ANTIHERMITIAN} \\ \{F, G\} \text{ is HERMITIAN} \end{array} \right] \Leftarrow \begin{array}{l} \text{WHEN } F \text{ IS HERMITIAN} \\ \text{AND } G \text{ IS HERMITIAN} \end{array} \quad \begin{array}{l} F = F^\dagger \\ G = G^\dagger \end{array}$$

$$\begin{aligned} \langle \psi | FG | \psi \rangle &= \langle \psi | \frac{1}{2} [F, G] + \frac{1}{2} \{F, G\} | \psi \rangle \\ &= \frac{1}{2} \langle \psi | [F, G] | \psi \rangle + \frac{1}{2} \langle \psi | \{F, G\} | \psi \rangle \\ &= \frac{1}{2} \underbrace{\langle [F, G] \rangle}_{\in \mathbb{I}_m} + \frac{1}{2} \underbrace{\langle \{F, G\} \rangle}_{\in \mathbb{R}} \\ &\quad \downarrow \text{IMAGINARY} \\ &= \mathbb{I}_m \langle \psi | FG | \psi \rangle + \mathbb{R} \langle \psi | FG | \psi \rangle \end{aligned}$$

$$|\langle FG \rangle|^2 \geq \frac{1}{4} |\langle \psi | [FG] | \psi \rangle|^2$$

$$\Rightarrow |\langle \psi | FG | \psi \rangle|^2 \geq \frac{1}{4} |\langle \psi | [FG] | \psi \rangle|^2$$

$$\langle \Delta A^2 \rangle \langle \Delta B^2 \rangle \geq \frac{1}{4} |\langle \psi | [A, B] | \psi \rangle|^2$$

REMEMBER: IF $z = a + ib$
 $|z|^2 = |a|^2 + |b|^2$
 $|z|^2 \geq |a|^2$

GENERAL UNCERTAINTY PRINCIPLE: SUPPOSE A AND B ARE TWO OBSERVABLES ON A QUANTUM SYSTEM AND LET K BE THE OBSERVABLE SUCH THAT

$$\begin{aligned} [A, B] &= iK \\ \text{THE, FOR ANY QUANTUM STATE } |\psi\rangle \\ \Delta A \Delta B &\geq \frac{1}{2} |\langle K \rangle| \end{aligned}$$

↑
 IF I GET INFORMATION FROM A I WILL LOSE INFO FROM B

Example

Problem 4.4 For a spin-1/2 particle, consider the generic state

$$|\psi\rangle = e^{i\phi} \cos\theta |z_+\rangle + e^{-i\phi} \sin\theta |z_-\rangle \quad \leftarrow \text{GENERIC STATE}$$

Find ΔS_x , ΔS_y , show that the spin component uncertainty relation in holds.

SHOW THAT THE HEISENBERG PRINCIPLE WORKS

$$\text{SHOW THAT } \Delta S_x \Delta S_y \geq \frac{1}{2} |\langle [S_x, S_y] \rangle| \quad \text{OR} \quad \Delta S_x^2 \Delta S_y^2 \geq \frac{1}{4} |\langle [S_x, S_y] \rangle|^2$$

$$\mathcal{B} = \{ |z_+\rangle, |z_-\rangle \}$$

$$\text{COMMUTATOR } [S_x, S_y] = i\hbar S_z$$

$$\Delta S_x^2 \Delta S_y^2 \geq \frac{1}{4} \hbar^2 |\langle S_z \rangle|^2$$

$$\hookrightarrow \langle S_z \rangle = \begin{pmatrix} e^{i\phi} \cos\theta \\ e^{-i\phi} \sin\theta \end{pmatrix}$$

$|\psi\rangle$

$$S_z = \frac{\hbar}{2} \sigma_z \Rightarrow \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\langle S_z \rangle = \langle \psi | S_z | \psi \rangle$$

$$\langle S_z \rangle = \langle \psi | S_z | \psi \rangle = (e^{-i\phi} \cos\theta, e^{i\phi} \sin\theta) \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} e^{i\phi} \cos\theta \\ e^{-i\phi} \sin\theta \end{pmatrix}$$

$$= \frac{\hbar}{2} (e^{-i\phi} \cos\theta, e^{i\phi} \sin\theta) \begin{pmatrix} e^{i\phi} \cos\theta \\ -e^{-i\phi} \sin\theta \end{pmatrix} = \frac{\hbar}{2} (\cos^2\theta - \sin^2\theta) = \frac{\hbar}{2} \cos 2\theta$$

$$\Delta S_x^2 = \langle S_x^2 \rangle - \langle S_x \rangle^2$$

$$S_x^2 = \left(\frac{\hbar}{2} x\right)^2 = \frac{\hbar^2}{4} x^2 = \frac{\hbar^2}{4} \mathbb{1}$$

$$x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \rightarrow x^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{1}$$

$$\langle S_x \rangle = \langle \psi | S_x | \psi \rangle = \langle \psi | \frac{\hbar}{4} \mathbb{1} | \psi \rangle = \frac{\hbar}{4} \langle \psi | \psi \rangle = \frac{\hbar}{4}$$

$$\begin{aligned} \langle \psi | S_x | \psi \rangle &= (e^{-i\phi} \cos \theta, e^{i\phi} \sin \theta) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{i\phi} \cos \theta \\ e^{-i\phi} \sin \theta \end{pmatrix} \frac{\hbar}{2} \\ &= \frac{\hbar}{2} (e^{-i\phi} \cos \theta, e^{i\phi} \sin \theta) \begin{pmatrix} e^{-i\phi} \sin \theta \\ e^{i\phi} \cos \theta \end{pmatrix} = \begin{pmatrix} e^{-2i\phi} \cos \theta \sin \theta \\ e^{2i\phi} \cos \theta \sin \theta \end{pmatrix} \frac{\hbar}{2} \end{aligned}$$

$$= \left(\frac{e^{-2i\phi} + e^{2i\phi}}{2} \right) \sin \theta \cos \theta \frac{\hbar}{2} = \cos(2\phi) \sin(2\theta) \frac{\hbar}{2}$$

$$\Delta S_x^2 = \langle S_x^2 \rangle - \langle S_x \rangle^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{4} \cos^2(2\phi) \sin^2(2\theta)$$

$$\Delta S_y^2 = \langle S_y^2 \rangle - \langle S_y \rangle^2$$

$$S_y = \frac{\hbar}{2} y \quad y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ y^2 = \mathbb{1}$$

$$\langle \psi | S_y | \psi \rangle = \frac{\hbar}{2} (e^{-i\phi} \cos \theta, e^{i\phi} \sin \theta) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} e^{i\phi} \cos \theta \\ e^{-i\phi} \sin \theta \end{pmatrix}$$

$$= \frac{\hbar}{2} (e^{-i\phi} \cos \theta, e^{i\phi} \sin \theta) \begin{pmatrix} -i e^{-i\phi} \sin \theta \\ i e^{i\phi} \cos \theta \end{pmatrix}$$

$$= \frac{\hbar}{2} i [-e^{-2i\phi} \cos \theta \sin \theta + e^{2i\phi} \cos \theta \sin \theta]$$

$$= \frac{\hbar}{2} 2i i \left[\frac{e^{2i\phi} - e^{-2i\phi}}{2i} \right] \cos \theta \sin \theta$$

$$= -\frac{\hbar}{2} \sin(2\theta) \sin 2\phi$$

$$\Delta S_y^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{4} \sin^2 2\theta \sin^2 2\phi$$

$$\Delta S_x^2 \Delta S_y^2 = \frac{\hbar^4}{4} (1 - \cos^2 2\phi \sin^2 2\theta) \frac{\hbar^4}{4} (1 - \sin^2 2\phi \sin^2 2\theta)$$

$$= \frac{\hbar^8}{16} \left[1 - \underbrace{\cos^2 2\phi \sin^2 2\theta - \sin^2 2\phi \sin^2 2\theta}_{-\sin^4 2\theta} + \sin^2 2\phi \cos^2 2\phi \sin^4 2\theta \right]$$

$$= \frac{\hbar^8}{16} \underbrace{\left[\cos^2 2\theta + \sin^4 2\theta \cos^2 2\phi \sin^4 2\theta \right]}_{\geq 0} = \frac{\hbar^2}{4} |\langle S_z \rangle|^2 = \frac{\hbar^2}{4} \cdot \frac{\hbar^2}{4} \cos^2 2\theta = \frac{\hbar^4}{16} \cos^4 2\theta$$